5	**(a)**	$E = I(R + r)$ $9.0 = 0.450(18 + r)$ $r = 2.0\ \Omega$	M1 A1
	(b)	$Q = It$ $= 0.450 \times 3600$ $= 1.6 \times 10^3 \text{C}$	M1 A1
	(c)	$E_{total} = IVt$ $= 0.450 \times 9.0 \times 3600$ $= 14.6 \times 10^3 \text{J}$	M1 A1
	(d)	$P = I^2 R$ $= (0.450)^2 \times 18.0$ $= 3.65\ \text{W}$	M1 A1
	(e)	New effective resistance $= \frac{V}{I} = \frac{9.0}{0.225} = 40.0\ \Omega$ Replacement resistance = 40.0 – 2.0 = 38.0 Ω	M1 A1

6	**(a)**	Magnetic field lines are directed <u>out of the paper</u>.	[A1]
	(b)	Magnetic force provides for centripetal force $Bqv = m\omega^2 r$ [M1] where ω is the angular velocity of the proton. $m\omega^2 r = Bq\omega r$ $\omega = \frac{Bq}{m}$ $\because \omega = \frac{2\pi}{T}$ [M1] where T is the period of the motion. $T = \frac{2\pi m}{Bq}$ [M1] $t_D = \frac{1}{2}T = \frac{\pi m}{Bq}$ [A0]	

		Alternative working: Magnetic force provides for centripetal force $Bqv = m\frac{v^2}{r}$ [M1] $r = \frac{mv}{Bq}$ [M1] $t_D = \frac{\pi r}{v}$ [M1] $= \frac{\pi m}{Bq}$ [A0]	
	(c)	The magnetic field does not change the speed of the proton. The electric field causes the speed to increase.	[B1] [B1]
	(d)	t_D or $\frac{\pi m}{Bq}$	[A1]
	(e)	Since the charge of the new particle is negative: This particle will make an anticlockwise direction motion within the dees. [B1] Since the mass-to-charge ratio (m/q) of the new particle is the same: This particle will make the same radius within the dees [B1] Since m/q of this new particle is the same: Based on the equation of **(b)** the t_D value will remain same since B remains the same. [B1]	

7	**(a)**	Nuclear fission is the <u>splitting / disintegration of a heavy nucleus into two lighter nuclei with approximately equal masses</u>.		A1
	(b)	**(i)**	3 neutrons	A1
	(b)	**(ii)**	Mass difference including neutrons = (234.993467 – 89.886883 – 142.905749 – 2 x 1.008665) x u = 0.183505 u [M1] Energy released = 0.183505 u c^2 = 2.742 x 10^{-11} J = 2.742 x 10^{-11} / (1.6 x 10^{-19}) = 171.375 [M1] = 171 MeV [A0]	
	(c)	**(i)**	Energy required from fission per second = 500 x 10^6 / 0.33 = 1.515 x 10^9 J	C1

			Number of fissions per second required = $1.515 \times 10^{9} / (2.742 \times 10^{-11})$ $= 5.53 \times 10^{19}$	A1
	(c)	**(ii)**	Mass of U-235 undergoing fission per second $= 5.53 \times 10^{19} \times 234.99$ u $= 2.156 \times 10^{-5}$ kg [C1] Mass of U-235 that fissions per day $= 2.156 \times 10^{-5} \times 24 \times 3600 = 1.863 = 1.9$ kg [A1]	
	(e)	Any two of the following: 1. Only a small amount of uranium is required to produce a large amount of electrical energy. This could remove Singapore's reliance on importing large amounts of fossil fuels. 2. The use of nuclear energy is carbon neutral or less polluting unlike the use of fossil fuels. Therefore this compares well with other carbon-neutral alternative energy sources. 3. Nuclear power can potentially generate large amounts of energy to meet Singapore's needs. 4. Nuclear power requires less land area than other alternative energy sources such as solar or wind power. This is suitable for land-scarce Singapore. 5. Nuclear power is more reliable (can be generated 24 hours a day) than other alternative energy sources such as wind or solar which depend on weather conditions. or other reasonable explanations.		B2

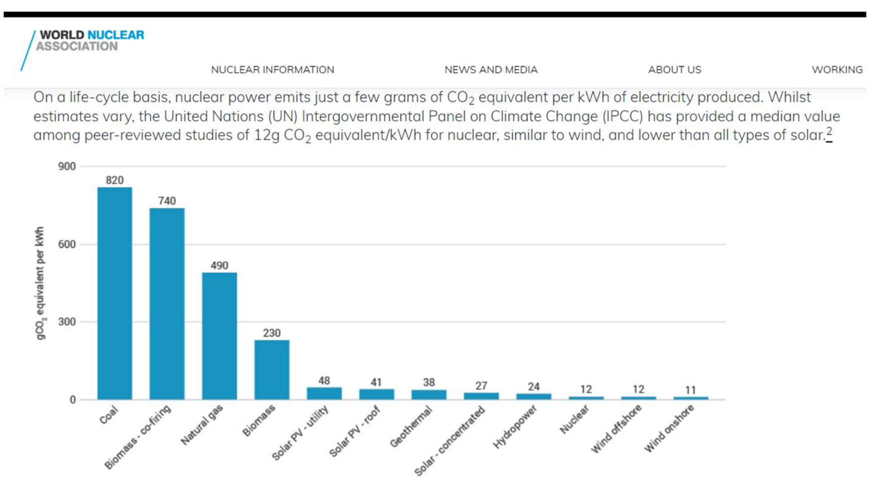

Figure 1: Average life-cycle CO_2 equivalent emissions (source: IPCC)

8	(a)	density of gas cloud = $1.66\times10^{-27}\times100=1.66\times10^{-25}$ kg cm^{-3} $=1.66\times10^{-19}$ kg m^{-3} [C1 if number of molecules obtained as 6.02 x 10^{59}] Volume of gas cloud = $\dfrac{10^{33}}{1.66\times10^{-19}}=6.02\times10^{51}$ m^3 Radius of spherical gas cloud: $\dfrac{4}{3}\pi R^3=6.02\times10^{51}$ $R=1.129\times10^{17}$ m $=1.1\times10^{17}$ m	C1 M1 A0
	(b)	Number of molecules, N = $\dfrac{10^{33}}{1.66\times10^{-27}}=6.02\times10^{59}$ Temperature T: $\dfrac{6.67\times10^{-11}\times\left(10^{33}\right)^2}{1.1\times10^{17}}\geq\dfrac{3}{2}\left(6.02\times10^{59}\right)\left(1.38\times10^{-23}\right)T$ T = 43.7 K (if used 1.1 x 10^{17}) T = 47.4 K (if used 1.129 x 10^{17}) T = 48.6 K (if used more dp) T = 52.5 K (if used 1.1 x 10^{17} to find volume)	C1 A1
	(c)	As the density of the gas cloud increases, <u>the mass of the gas cloud increases</u> since the volume stays the same. (also accept <u>number of molecules increases</u>) From the formula, the maximum (average) <u>temperature</u> for the gas cloud to collapse will also <u>increase</u>. (Since both M and N increases, but M^2 increases more than N, thus the ratio $\dfrac{M^2}{N}$ increases).	M1 A1
	(d)	High kinetic energy is required to <u>overcome the (mutual electrostatic) repulsion</u> between the positively charged nuclei so that they can fuse. Or nuclei must <u>have high speed</u> to collide. Since thus <u>high kinetic energy</u> (of the nuclei) is related to temperature, thus a high temperature is required for fusion.	B1 B1

	(e)	Energy released = total binding energy of products – total binding energy of reactants Energy released = (7.07 x 4) – (2.57 x 3 x 2) = 12.86 MeV = 12.9 MeV	 C1 A1
	(f)	The maximum frequency gamma photon is emitted when all the energy released in the fusion reaction is converted into the photon energy. hf = 12.9 MeV $f = \dfrac{12.9\times10^{6}\times1.6\times10^{-19}}{6.63\times10^{-34}}$ $= 3.11\times10^{21}$ Hz	 C1 A1
	(g)	Iron has the <u>highest binding energy per nucleon</u> of any element Or Iron is the <u>most stable element</u> Hence, the fusion of iron into heavier elements<u> requires the input of energy</u> / is <u>unlikely to occur spontaneously</u>. Or There is <u>not enough energy / temperature</u> in the stars for fusion of iron.	B1 B1
	(h)	gravitational force provides for centripetal force $\dfrac{GMm}{r^2} = \dfrac{mv^2}{r}$ $v = \sqrt{\dfrac{GM}{r}}$	C1 M1 A0
	(i)	$v = \sqrt{\dfrac{GM}{r}}$ $v = \sqrt{\dfrac{G\left(\frac{4}{3}\pi r^3 \rho\right)}{r}}$ $v = \sqrt{\dfrac{4}{3}G\pi\rho} \times r$ <u>Since</u> G, π, ρ <u>are constants</u>, so $v \propto r$	 C1 M1 A0

	(j)	The mass is uniformly distributed in a sphere nearer the galactic centre and increases proportionally with distance further from the galactic centre. **OR** Messier 33 is likely to have a spherical distribution of mass with uniform density nearer the galactic centre, and non-uniform density further from the galactic centre. (This is because the portion of the rotation curve nearer M33's core shows a proportional relationship, similar to Fig 8.3(b) while the portion of the rotation curve further from M33's core shows an almost horizontal relationship, similar to Fig 8.3(c).)	B1 B1